American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Intermediate Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

+ − * 2 3 / 4 2 / * 8 + 1 3 4

A. 4
B. 6
C. 8
D. 10
E. 12

2. Prefix/Infix/Postfix Notation

Translate the following infix expression into postfix:

$$\frac{xy + z}{a^2}$$

A. x y z * + a 2 ^ /
B. x y * + z a 2 ^ /
C. x y * z + a 2 ^ /
D. x * y z + a 2 ^ /
E. x y * z a 2 ^ + /

3. Bit-String Flicking

Evaluate:

10110 OR 11011 AND 00101 XOR 01101

A. 11110
B. 00011
C. 01110
D. 11011
E. 01000

4. Bit-String Flicking

Solve the following equation for X if X is a 5-bit string:

(NOT (RCIRC-3 (LSHIFT-2 X))) = 01100

A. **101
B. 1*110
C. *1000
D. **111
E. 0*011

5. LISP

Evaluate the following LISP expression:

(ADD 2 (SUB 7 3) 3 (MULT 2 5) 8)

A. 11
B. 13
C. 23
D. 25
E. 27

6. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR (CDR '(e (f g) (d b (a c)))))))

A. (d b (a c))
B. (b (a c))
C. (f g)
D. (b)
E. (e (f g))

American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Intermediate Division

1. 前缀表达式/中缀表达式/后缀表达式

如果下方所有数字都是一位数，请计算下方前缀表达式：:

+ − * 2 3 / 4 2 / * 8 + 1 3 4

A. 4
B. 6
C. 8
D. 10
E. 12

2. 前缀表达式/中缀表达式/后缀表达式

请将下方中缀表达式转换为后缀表达式：

$$\frac{xy + z}{a^2}$$

A. x y z * + a 2 ^ /
B. x y * + z a 2 ^ /
C. x y * z + a 2 ^ /
D. x * y z + a 2 ^ /
E. x y * z a 2 ^ + /

3. 比特位操作

请计算下方表达式：

10110 OR 11011 AND 00101 XOR 01101

A. 11110
B. 00011
C. 01110
D. 11011
E. 01000

4. 比特位操作

如果 X 是一个 5 位的位串，请根据下列方程求 X 的值：

(NOT (RCIRC-3 (LSHIFT-2 X))) = 01100

A. **101
B. 1*110
C. *1000
D. **111
E. 0*011

5. LISP 语言

请计算下方 LISP 表达式：

(ADD 2 (SUB 7 3) 3 (MULT 2 5) 8)

A. 11
B. 13
C. 23
D. 25
E. 27

6. LISP 语言

请计算下方 LISP 表达式：

(CDR (CAR (CDR (CDR '(e (f g) (d b (a c)))))))

A. (d b (a c))
B. (b (a c))
C. (f g)
D. (b)
E. (e (f g))